Spectroscopic measurement of the energy gap in Ge and Si samples

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Abstract

The energy gap of Si and Ge semiconductors are measured, investigating photoconductivity and transmissivity in the infrared-visible range. The light is divided thanks to a rotating diffraction grating. The estimated values of the energy gap are obtained averaging for clockwise/anticlockwise rotation of the grating, for positive/negative angles, for photoconductivity/transmissivity signals; in order to cancel out sistematic errors. For Ge, $E_g = (0.65 \pm 0.02)$ eV (0.66 eV literature value) and for Si $E_g = (1.1 \pm 0.1)$ eV (1.12 literature value).

Introduction

Semiconductor are characterized by an energy gap E_g, which separates valence from conduction band. This means that photons start to be adsorbed when their energy is greater than the energy gap. We then expect that the absorption coefficient increase. Another way to probe energy gap is to measure conductivity of the semiconducting sample: photon adsorption generates electron-hole free carriers and therefore also the conductivity increases.

Materials and Methods

Measures of conductivity are simply obtained by biasing the sample with a constant current and measuring the voltage drop. In this experiment, a pyroelectric sensor is used to measured the transmission spectra. The apparatus is composed by a "V" shaped bench of semiangle $\phi = 7.92°$, a diffraction grating [1] is placed at the vertex. The grating decomposes the spectrum of the white light and is rotated automatically from -50° to 50°. A lens placed near the sample focuses the light and a slit selects a narrow range of wavelength for each angle. From the grating equation, we can convert the incident angle θ to wavelength, and then energy: $E = hc/2a \cos\phi \sin\theta$, where a is the grating constant, equal to 1.667 μm. The signals are amplified by a twin-channel lock-in, for this reason the light chopped with a certain constant frequency. In order to collect a good signal, AC and DC gain can be changed directly from the control unit. AC gain increases the signal that comes in the amplifier, DC gain increases the output signal. The two samples are crystalline Ge and Si wafer, with thickness of the order of μm. The rotary sensor, responsable of the rotation of the grating, needs a calibration: an angle is associated at each voltage signal from the sensor.

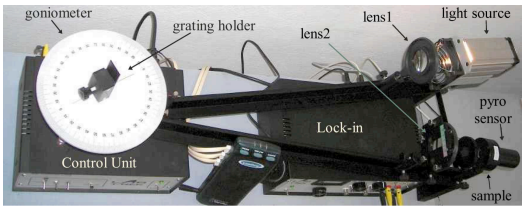


Figure 1: Experimental apparatus

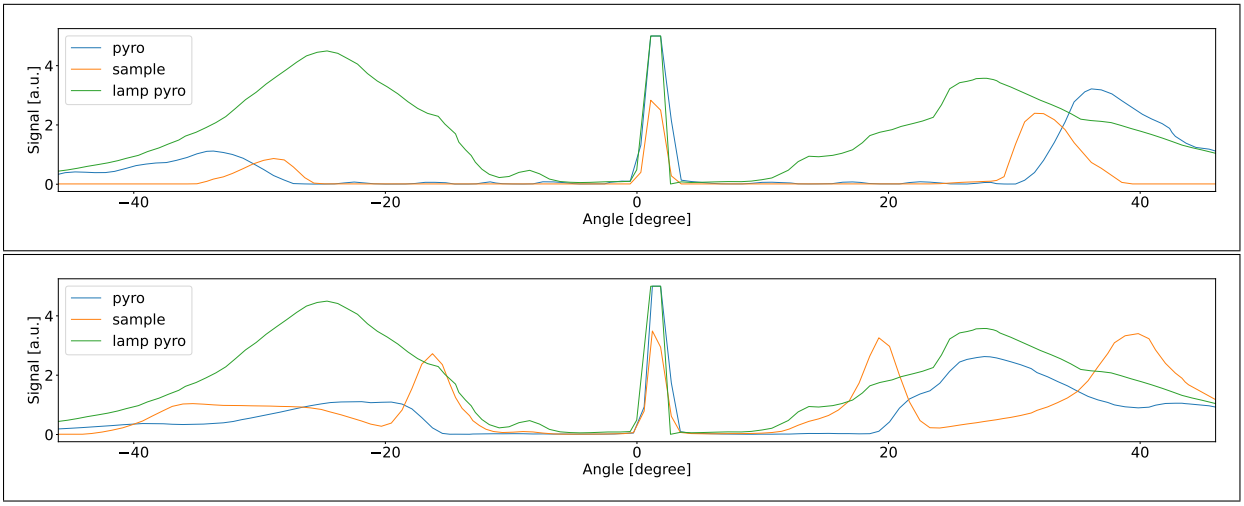


Figure 2: Signals from CW acquisition. Both Ge (upper) and Si (lower) have a peak at 0°. This represents the zero'th order solution of the grating's equation. The second order peak is found only in Si, the associated wavelenght is 2λ.

Results and Discussion

For each sample, a clockwise (CW) and counter-clockwise (CCW) acquisition is performed. Fig. (2) illustrates Ge and Si spectra for the CW acquisition. Transmission coefficient is expected to be constant before E_g. This coefficient is not proportional to pyroelectric signal, because the lamp does not emit with a consant spectral distribution. The normalized spectrum is obtained by dividing the pyro signal of the semiconductor by the pyro signal of the light (without placing the sample). The spectra need to be interpolated, in order to acquire the same value on the theta (or energy) axis. There are a lot of effects that influence the position of the energy gap and the rising/falling edge of the two signals: impurity states and surface states between the energy gap are responsible of absorbtion of photons with energies below E_g. Moreover, the spectra are not symmetric for positive and negative angles: for this reason we make an average of the values of E_g fow CW and CCW acquisition and for positive and negative angles. This results in 8 guesses for each sample, the uncertanty is taken as the standard deviation between theese values. With reference to Fig. (3), we make the assumption that rising and falling edges are linear; the estimated value of E_g is the x-value of the intersection point between the constant line below E_g and the raising/falling linear edge.

Conclusions

The calculated values of the energy gap are (0.65 ± 0.02) eV for Ge and (1.1 ± 0.1) eV for Si. This values are in agreement with the expected E_g, which are, respectively, 0.66 eV and 1.12 eV [2].

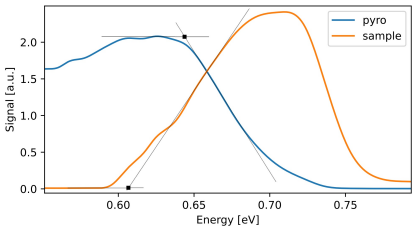


Figure 3: Estimation of E_g (the two black squares)

References

[1] *600 Grooves, 25mm Square, 750nm Ruled Diffraction Grating.* URL: https://www.edmundoptics.com/p/600-grooves-25mm-square-750nm-ruled-diffraction-grating/4658/.

[2] Ben G. Streetman. *Solid State Electronic Devices*. Prentice-Hall, Inc., 1990.